Example 12 Using Remark 1, by the formula for integration by parts in an improper integral, we find that

$$\int_{\pi/2}^{+\infty} \frac{\sin x}{x}\,\mathrm{d}x = -\frac{\cos x}{x}\bigg|_{\pi/2}^{+\infty} - \int_{\pi/2}^{+\infty} \frac{\cos x}{x^2}\,\mathrm{d}x = -\int_{\pi/2}^{+\infty} \frac{\cos x}{x^2}\,\mathrm{d}x,$$

provided the last integral converges. But, as we saw in Example 5, this integral converges, and hence the integral

$$\int_{\pi/2}^{+\infty} \frac{\sin x}{x}\,\mathrm{d}x \tag{6.77}$$

also converges.

At the same time, the integral (6.77) is not absolutely convergent. Indeed, for $b \in [\pi/2, +\infty[$ we have

$$\int_{\pi/2}^{b} \left|\frac{\sin x}{x}\right| \mathrm{d}x \geq \int_{\pi/2}^{b} \frac{\sin^2 x}{x}\,\mathrm{d}x = \frac{1}{2}\int_{\pi/2}^{b} \frac{\mathrm{d}x}{x} - \frac{1}{2}\int_{\pi/2}^{b} \frac{\cos 2x}{x}\,\mathrm{d}x. \tag{6.78}$$

The integral

$$\int_{\pi/2}^{+\infty} \frac{\cos 2x}{x}\,\mathrm{d}x,$$

as can be verified through integration by parts, is convergent, so that as $b \to +\infty$, the difference on the right-hand side of relation (6.78) tends to $+\infty$. Thus, by estimate (6.78), the integral (6.77) is not absolutely convergent.

We now give a special convergence test for improper integrals based on the second mean-value theorem and hence essentially on the same formula for integration by parts.

Proposition 4 (Abel–Dirichlet test for convergence of an integral) *Let $x \mapsto f(x)$ and $x \mapsto g(x)$ be functions defined on an interval $[a, \omega[$ and integrable on every closed interval $[a, b] \subset [a, \omega[$. Suppose that g is monotonic.*

Then a sufficient condition for convergence of the improper integral

$$\int_a^{\omega} (f \cdot g)(x)\,\mathrm{d}x \tag{6.79}$$

is that the one of the following pairs of conditions hold:

α_1) *the integral $\int_a^{\omega} f(x)\,\mathrm{d}x$ converges,*
β_1) *the function g bounded on $[a, \omega[$,*

or

α_2) *the function $\mathcal{F}(b) = \int_a^b f(x)\,\mathrm{d}x$ is bounded on $[a, \omega[$,*
β_2) *the function $g(x)$ tends to zero as $x \to \omega$, $x \in [a, \omega[$.*

Proof For any b_1 and b_2 in $[a, \omega[$ we have, by the second mean-value theorem,

$$\int_{b_1}^{b_2} (f \cdot g)(x)\,\mathrm{d}x = g(b_1)\int_{b_1}^{\xi} f(x)\,\mathrm{d}x + g(b_2)\int_{\xi}^{b_2} f(x)\,\mathrm{d}x,$$

where ξ is a point lying between b_1 and b_2. Hence by the Cauchy convergence criterion (Proposition 2), we conclude that the integral (6.79) does indeed converge if either of the two pairs of conditions holds. □

6.5.3 Improper Integrals with More than One Singularity

Up to now we have spoken only of improper integrals with one singularity caused either by the unboundedness of the function at one of the endpoints of the interval of integration or by an infinite limit of integration. In this subsection we shall show in what sense other possible variants of an improper integral can be taken.

If both limits of integration are singularities of either of these two types, then by definition

$$\int_{\omega_1}^{\omega_2} f(x)\,\mathrm{d}x := \int_{\omega_1}^{c} f(x)\,\mathrm{d}x + \int_{c}^{\omega_2} f(x)\,\mathrm{d}x, \tag{6.80}$$

where c is an arbitrary point of the open interval $]\omega_1, \omega_2[$.

It is assumed here that each of the improper integrals on the right-hand side of (6.80) converges. Otherwise we say that the integral on the left-hand side of (6.80) diverges.

By Remark 2 and the additive property of the improper integral, the definition (6.80) is unambiguous in the sense that it is independent of the choice of the point $c \in]\omega_1, \omega_2[$.

Example 13

$$\int_{-1}^{1} \frac{\mathrm{d}x}{\sqrt{1-x^2}} = \int_{-1}^{0} \frac{\mathrm{d}x}{\sqrt{1-x^2}} + \int_{0}^{1} \frac{\mathrm{d}x}{\sqrt{1-x^2}} =$$
$$= \arcsin x\big|_{-1}^{0} + \arcsin x\big|_{0}^{1} = \arcsin x\big|_{-1}^{1} = \pi.$$

Example 14 The integral

$$\int_{-\infty}^{+\infty} \mathrm{e}^{-x^2}\,\mathrm{d}x$$

is called the *Euler–Poisson integral*, and sometimes the *Gaussian integral*. It obviously converges in the sense given above. It will be shown later that its value is $\sqrt{\pi}$.

continuous as a function on $\mathbb{R}$, then the new function $(x, y) \overset{F}{\longmapsto} f(x)$ will be continuous as a function on $\mathbb{R}^2$. This can be verified either directly from the definition of continuity or by remarking that the function F is the composition $(f \circ \pi^1)(x, y)$ of continuous functions.

In particular, it follows from this, when we take account of c) and e), that the functions

$$f(x, y) = \sin x + e^{xy}, \qquad f(x, y) = \arctan\big(\ln\big(|x| + |y| + 1\big)\big),$$

for example, are continuous on $\mathbb{R}^2$.

We remark that the reasoning just used is essentially local, and the fact that the functions f and F studied in Example 7 were defined on the entire real line $\mathbb{R}$ or the plane $\mathbb{R}^2$ respectively was purely an accidental circumstance.

Example 8 The function $f(x, y)$ of Example 2 is continuous at any point of the space $\mathbb{R}^2$ except $(0, 0)$. We remark that, despite the discontinuity of $f(x, y)$ at this point, the function is continuous in either of its two variables for each fixed value of the other variable.

Example 9 If a function $f : E \to \mathbb{R}^n$ is continuous on the set E and $\tilde{E}$ is a subset of E, then the restriction $f|_{\tilde{E}}$ of f to this subset is continuous on $\tilde{E}$, as follows immediately from the definition of continuity of a function at a point.

We now turn to the global properties of continuous functions. To state them for functions $f : E \to \mathbb{R}^n$, we first give two definitions.

Definition 8 A mapping $f : E \to \mathbb{R}^n$ of a set $E \subset \mathbb{R}^m$ into $\mathbb{R}^n$ is *uniformly continuous* on E if for every $\varepsilon > 0$ there is a number $\delta > 0$ such that $d(f(x_1), f(x_2)) < \varepsilon$ for any points $x_1, x_2 \in E$ such that $d(x_1, x_2) < \delta$.

As before, the distances $d(x_1, x_2)$ and $d(f(x_1), f(x_2))$ are assumed to be measured in $\mathbb{R}^m$ and $\mathbb{R}^n$ respectively.

When $m = n = 1$, this definition is the definition of uniform continuity of numerical-valued functions that we already know.

Definition 9 A set $E \subset \mathbb{R}^m$ is *pathwise connected* if for any pair of its points x_0, x_1, there exists a path $\Gamma : I \to E$ with support in E and endpoints at these points.

In other words, it is possible to go from any point $x_0 \in E$ to any other point $x_1 \in E$ without leaving E.

Since we shall not be considering any other concept of connectedness for a set except pathwise connectedness for the time being, for the sake of brevity we shall temporarily call pathwise connected sets simply *connected*.

Definition 10 A *domain* in $\mathbb{R}^m$ is an open connected set.

Example 10 An open ball $B(a;r)$, $r>0$, in $\mathbb{R}^m$ is a domain. We already know that $B(a;r)$ is open in $\mathbb{R}^m$. Let us verify that the ball is connected. Let $x_0=(x_0^1\dots,x_0^m)$ and $x_1=(x_1^1,\dots,x_1^m)$ be two points of the ball. The path defined by the functions $x^i(t)=tx_1^i+(1-t)x_0^i$ $(i=1,\dots,m)$, defined on the closed interval $0\le t\le 1$, has x_0 and x_1 as its endpoints. In addition, its support lies in the ball $B(a;r)$, since, by Minkowski's inequality, for any $t\in[0,1]$,

$$d\big(x(t),a\big)=\sqrt{\sum_{i=1}^m\big(x^i(t)-a^i\big)^2}=\sqrt{\sum_{i=1}^m\big(t\big(x_1^i-a^i\big)+(1-t)\big(x_0^i-a^i\big)\big)^2}\le$$

$$\le\sqrt{\sum_{i=1}^m\big(t\big(x_1^i-a^i\big)\big)^2}+\sqrt{\sum_{i=1}^m\big((1-t)\big(x_0^i-a^i\big)\big)^2}=$$

$$=t\cdot\sqrt{\sum_{i=1}^m\big(x_1^i-a^i\big)^2}+(1-t)\cdot\sqrt{\sum_{i=1}^m\big(x_0^i-a^i\big)^2}<tr+(1-t)r=r.$$

Example 11 The circle (one-dimensional sphere) of radius $r>0$ is the subset of $\mathbb{R}^2$ given by the equation $(x^1)^2+(x^2)^2=r^2$. Setting $x^1=r\cos t$, $x^2=r\sin t$, we see that any two points of the circle can be joined by a path that goes along the circle. Hence a circle is a connected set. However, this set is not a domain in $\mathbb{R}^2$, since it is not open in $\mathbb{R}^2$.

We now state the basic facts about continuous functions in the large.

Global Properties of Continuous Functions

a) *If a mapping $f:K\to\mathbb{R}^n$ is continuous on a compact set $K\subset\mathbb{R}^m$, then it is uniformly continuous on K.*

b) *If a mapping $f:K\to\mathbb{R}^n$ is continuous on a compact set $K\subset\mathbb{R}^m$, then it is bounded on K.*

c) *If a function $f:K\to\mathbb{R}$ is continuous on a compact set $K\subset\mathbb{R}^m$, then it assumes its maximal and minimal values at some points of K.*

d) *If a function $f:E\to\mathbb{R}$ is continuous on a connected set E and assumes the values $f(a)=A$ and $f(b)=B$ at points $a,b\in E$, then for any C between A and B, there is a point $c\in E$ at which $f(c)=C$.*

Earlier (Sect. 4.2), when we were studying the local and global properties of functions of one variable, we gave proofs of these properties that extend to the more general case considered here. The only change that must be made in the earlier proofs is that expressions of the type $|x_1-x_2|$ or $|f(x_1)-f(x_2)|$ must be replaced by $d(x_1,x_2)$ and $d(f(x_1),f(x_2))$, where d is the metric in the space where the points in question are located. This remark applies fully to everything except the last statement d), whose proof we now give.

Proof d) Let $\Gamma : I \to E$ be a path that is a continuous mapping of an interval $[\alpha, \beta] = I \subset R$ such that $\Gamma(\alpha) = a$, $\Gamma(\beta) = b$. By the connectedness of E there exists such a path. The function $f \circ \Gamma : I \to \mathbb{R}$, being the composition of continuous functions, is continuous; therefore there is a point $\gamma \in [\alpha, \beta]$ on the closed interval $[\alpha, \beta]$ at which $f \circ \Gamma(\gamma) = C$. Set $c = \Gamma(\gamma)$. Then $c \in E$ and $f(c) = C$. □

Example 12 The sphere $S(0; r)$ defined in $\mathbb{R}^m$ by the equation

$$\left(x^1\right)^2 + \cdots + \left(x^m\right)^2 = r^2,$$

is a compact set.

Indeed, it follows from the continuity of the function

$$\left(x^1, \ldots, x^m\right) \mapsto \left(x^1\right)^2 + \cdots + \left(x^m\right)^2$$

that the sphere is closed, and from the fact that $|x^i| \le r$ $(i = 1, \ldots, m)$ on the sphere that it is bounded.

The function

$$\left(x^1, \ldots, x^m\right) \mapsto \left(x^1\right)^2 + \cdots + \left(x^k\right)^2 - \left(x^{k+1}\right)^2 - \cdots - \left(x^m\right)^2$$

is continuous on all of $\mathbb{R}^m$, so that its restriction to the sphere is also continuous, and by the global property c) of continuous functions assumes its minimal and maximal values on the sphere. At the points $(1, 0, \ldots, 0)$ and $(0, \ldots, 0, 1)$ this function assumes the values 1 and -1 respectively. By the connectedness of the sphere (see Problem 3 at the end of this section), global property d) of continuous functions now enables us to assert that there is a point on the sphere where this function assumes the value 0.

Example 13 The open set $\mathbb{R}^m \backslash S(0; r)$ for $r > 0$ is not a domain, since it is not connected.

Indeed, if $\Gamma : I \to \mathbb{R}^m$ is a path one end of which is at the point $x_0 = (0, \ldots, 0)$ and the other at some point $x_1 = (x_1^1, \ldots, x_1^m)$ such that $(x_1^1)^2 + \cdots + (x_1^m)^2 > r^2$, then the composition of the continuous functions $\Gamma : I \to \mathbb{R}^m$ and $f : \mathbb{R}^m \to \mathbb{R}$, where

$$\left(x^1, \ldots, x^m\right) \xmapsto{f} \left(x^1\right)^2 + \cdots + \left(x^m\right)^2,$$

is a continuous function on I assuming values less than r^2 at one endpoint and greater than r^2 at the other. Hence there is a point γ on I at which $(f \circ \Gamma)(\gamma) = r^2$. Then the point $x_\gamma = \Gamma(\gamma)$ in the support of the path turns out to lie on the sphere $S(0; r)$. We have thus shown that it is impossible to get out of the ball $B(0; r) \subset \mathbb{R}^m$ without intersecting its boundary sphere $S(0; r)$.